Aubrianne Thoms

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EDUCATION

Randolph-Macon College, Ashland, VA — September 2021 – May 2025
Bachelor of Science in Physics, Minors in Astrophysics, Mathematics, and Asian Studies
3.36 on 4.0 scale

Florida Institute of Technology, Melbourne, FL — August 2025 - Present
Master of Science in Space Sciences
3.0 on 4.0 scale

SKILLS

- ***Beginner:*** Machine Learning (Decision Tree model, Naive Bayes Classifier, etc.), Kalibr, MATLAB
- ***Intermediate:*** Python (Google Colab, Spyder, Jupyter Notebook), LaTEX, AstroImageJ, Linux/Ubuntu, Cameras (D455 RealSense, CCDs, RSpecExplorer Spectrometer), Physics Lab Equipment (Spectrometer, Leslie Cube, h/e apparatus), ROS 1, GCM (ExoPlaSim)
- ***Advanced:*** Telescope Operation & Maintenance

EXPERIENCE

Robotics Research Assistant — September 2025 – Present
Florida Institute of Technology; Melbourne, FL

- Objective is to improve localization programming for autonomous robots
- Responsible for refining camera calibration and AprilTag detection techniques
- Have used a USB camera and D455 RealSense sensor complete with a RGB stream, infrared stream, and an IMU

Lead Senior Student Observatory Assistant — September 2023 – May 2025
Randolph-Macon College; Ashland, VA

- Helped arrange public viewings and promoted events on social media
- Operated the telescope to observe night sky objects; an ASA 40 cm Ritchey-Chrétien F/8 model
- Utilized creative engagement opportunities, such as stickers and BINGO cards, for outreach

Research Experiences for Undergraduates (REU) Intern — June 2024 – August 2024
University of Texas at Rio Grande Valley; Brownsville, TX

- Funded by the National Science Foundation
- Used data of optical follow-up observations of gamma-ray bursts
- Worked on a Python pipeline for analysis; Goal was to create light curves

Schapiro Undergraduate Research Fellowship (SURF) Intern — June 2023 – August 2023
Randolph-Macon College; Ashland, VA

- Confirmed that Keeble Observatory can detect exoplanet transits
- Collected data using an ASA 40 cm Ritchey-Chrétien telescope equipped with a CCD
- Create transit curves through Python code and AstroImageJ

RESEARCH

- **Exoplanet General Circulation Model Research**

09/2025 - Present

- Investigating how aerosols behave in atmospheres of tidally-locked planets using three dimensional global circulation model *ExoPaSim*
- Pivoted to exploring storm climatology in exoplanet atmospheres for master's thesis

- **Optical Transient Research**
  09/2024 - 05/2025
  - Continuation of GRB follow-up observations and pipeline troubleshooting; collaboration with UTRGV
- **Exoplanet Transit Research**
  09/2023 - 05/2024
  - Follow-up observations of giant exoplanets orbiting M-dwarf stars; in collaboration with the *GEMS*

HONORS, CONFERENCES, AND PUBLICATION

- **Honors and awards:**
  - **George Spagna Jr. Award for Excellence in Astrophysics:** 2025; R-MC Department of Physics, Engineering, and Astrophysics
  - **The William Hesse Memorial Award:** 2024; R-MC Department of Physics, Engineering, and Astrophysics
  - **The Haun Physics Scholarship:** 2024-2025; R-MC Department of Physics, Engineering, and Astrophysics
  - **Delta Alpha Pi:** Randolph-Macon College; Vice President (2024-2025); Secretary (2023-2024)
- **Conferences and Talk:**
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  - **Florida Space Summit** - Kennedy Space Center - Poster Presentation - October 21-22, 2026
  - **23rd Annual Meeting of Asia Oceanic Geosciences Society** - Fukuoka, Japan - Poster Presentation - August 2-7, 2026
  - **American Association of Physics Teachers** - Jefferson Lab – Poster Presentation – October 2024
  - **CU*IP** - West Virginia University – Poster Presentation - January 2024
- **Collaborative Publication:**
  - ***Searching for GEMS: Characterizing Six Giant Planets Around Cool Dwarfs –*** The Astronomical Journal

EXTRACURRICULARS AND HOBBIES

- **Study Abroad:**
  - **England 2022**: Courses "Physics of Light" and "Psychology of Vision"; Wroxton and London
  - **Japan 2025**: Courses "Film Music in Japan" and "Comparative Legislatures"; Tokyo, Kyoto, and Hiroshima
- **Ortega Observatory Member:** January 2026 – Present
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  - **Social Media Team**
- **Student Government Association:** *Vice President of External Relations:* September 2024 - May 2025
- **Society of Physics Students:** *Vice President*: September 2024 - May 2025; *Secretary*: September 2023 - May 2024
- Instruments (Clarinet, Oboe, Guitar) – Crafting (Sewing, Collaging, Drawing) – Reading/Writing – Bug Pinning/Entomology – Fitness (Yoga, Rollerskating, etc.)